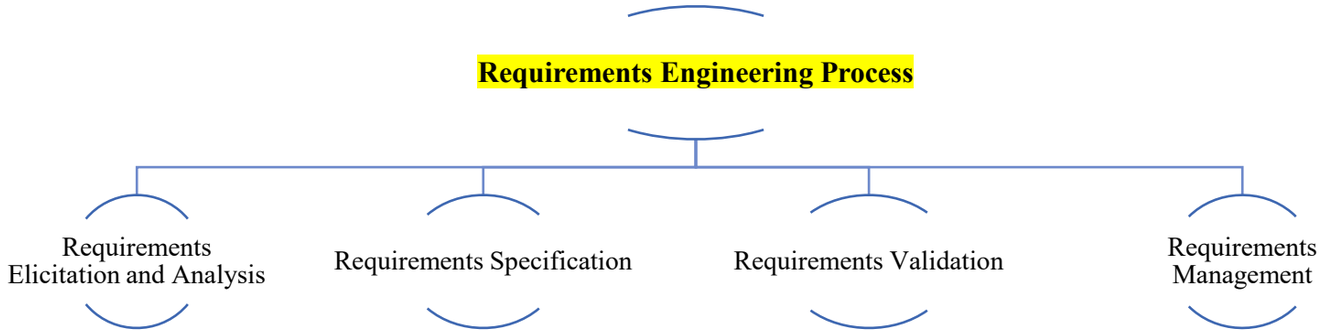


Step	Description	Example
Requirements Elicitation & Analysis	Collect requirements from stakeholders, analyze them, resolve conflicts, remove ambiguities, and prioritize them. Common techniques include interviews, questionnaires, observation, workshops, and prototyping.	Interview librarians, students, and administrators to determine required features such as searching books, borrowing, returning, and generating reports. Resolve conflicting requests and prioritize essential features.
Requirements Specification (SRS)	Document the agreed requirements in a structured Software Requirements Specification (SRS). The SRS clearly defines **what** the system should do, not **how** it will be implemented.	The SRS states requirements such as: *"The system shall allow students to search books by title, author, or ISBN."* It also documents performance, security, and usability requirements.
Requirements Validation	Review the SRS to ensure that the requirements are correct, complete, consistent, feasible, and testable. Stakeholders verify that the documented requirements reflect their actual needs.	Librarians review the SRS and confirm that all required library functions are included and that no important requirements are missing or contradictory.
Requirements Management	Manage changes to requirements throughout the software development lifecycle. New requirements are evaluated, documented, approved, and tracked to maintain consistency.	After deployment, the university requests a new feature that allows students to reserve books online. The new requirement is added to the SRS, assigned a unique identifier, and implemented in a future release.

Requirements Elicitation & Analysis

Techniques for collecting requirements

Technique	Purpose	Library Management System Example
Interviews	Ask stakeholders about their needs.	Interview librarians about book borrowing procedures.
Observation (Ethnography)	Observe users performing their daily tasks.	Watch librarians issue and return books.
Questionnaires	Collect information from many users quickly.	Survey students about desired library features.
Workshops	Gather stakeholders together to discuss requirements.	Conduct a meeting with librarians, students, and administrators.
Scenarios / Use Cases	Describe how users interact with the system.	"A student searches for a book and borrows it."
Prototyping	Build a simple model to gather feedback.	Show a prototype of the online book search page.
Document Analysis	Study existing forms, manuals, and reports.	Review current library borrowing registers and policies.

Requirements Validation Process

Validation Technique	Description	Library Management System Example
Requirements Review (Inspection)	Stakeholders review the SRS to identify missing, incorrect, or conflicting requirements.	Librarians review the SRS and notice that the "Book Reservation" feature is missing.
Prototyping	Develop a simple prototype or mock-up so users can evaluate the proposed system.	Show a prototype of the book search page. Students suggest adding filters by category and author.
Test Case Generation	Create test cases from the requirements to determine whether they are clear and testable.	Requirement: "Students can borrow up to 5 books." Test case: Try borrowing a 6th book; the system should reject it.
Consistency Checking	Verify that no requirements contradict each other.	One requirement says students can borrow 5 books, while another says 10 books. The conflict must be resolved.
Completeness Checking	Ensure that all required features, constraints, and business rules are included.	Verify that searching, borrowing, returning, fine calculation, and report generation are all documented.
Feasibility Checking	Confirm that each requirement is technically, economically, and operationally feasible.	A requirement for an AI-based voice assistant may exceed the project's budget and schedule.

Requirements Validation Process

i. Draft SRS
ii. Review Requirements
iii. Identify Errors & Conflicts
iv. Correct the Requirements
v. Approve Final SRS

Checklist for Validating Requirements

Question	Yes / No
Are the requirements correct?	✓
Are the requirements complete?	✓
Are the requirements consistent?	✓
Are the requirements unambiguous?	✓
Are the requirements feasible?	✓
Are the requirements testable?	✓
Have all stakeholders approved of them?	✓

Example: FR-01: The system shall allow students to borrow a maximum of 5 books at a time.

Validation Step	Result
Reviewed by librarian	✓ Approved
Reviewed by students	✓ Approved
Test case prepared	✓ Borrow 6th book → Rejected
Feasible	✓ Yes
Clear and measurable	✓ Yes
Final decision	**Validated**

Requirements Management

Definition: Requirements Management is the process of **documenting, tracking, controlling, and updating software requirements throughout the Software Development Life Cycle (SDLC).** Its main goal is to ensure that any change to a requirement is properly evaluated, approved, documented, and communicated to all stakeholders.

Requirements Management is the process of managing changes to software requirements throughout the project.

Requirements Management Process

Step	Description
Identify Change	A stakeholder requests a new requirement or modification.
Analyze Impact	Determine how the change affects cost, time, design, testing, and other requirements.
Approve or Reject	The project manager or change control board decides whether to accept the change.
Update Requirements	Modify the Software Requirements Specification (SRS) and related documents.
Implement the Change	Developers implement the approved requirement.
Verify the Change	Test the new functionality to ensure it works correctly.

Example: Library Management System

Original Requirement

FR-05: The system shall allow students to borrow a maximum of **5 books**.

New Requirement

The university changes its policy. Students can now borrow **7 books**.

Requirements Management Process	
Step	**Example**
Change Request	Library administration requests increasing the borrowing limit.
Impact Analysis	Developers identify changes needed in the database, borrowing module, and test cases.
Approval	Project manager approves the change.
Update SRS	FR-05 is updated from **5 books** to **7 books**.
Implementation	Developers modify the borrowing logic.
Testing	Testers verify that students can borrow up to 7 books but not an 8th book.

Requirement Traceability

One important activity in Requirements Management is **traceability**.

Traceability means tracking each requirement throughout the SDLC

Requirement ID	Requirement	Design Module	Code Module	Test Case
FR-01	User Login	Login Module	Login.java	TC-01
FR-02	Search Books	Search Module	Search.java	TC-02
FR-03	Borrow Books	Borrow Module	Borrow.java	TC-03

Version Control of Requirements

Requirements change over time.

Version	Date	Change
V1.0	01 Aug 2026	Initial SRS
V1.1	15 Aug 2026	Added Email Notification
V1.2	25 Aug 2026	Borrowing limit changed from 5 to 7 books
V2.0	10 Sep 2026	Added QR Code Borrowing

Version control ensures everyone works with the latest approved requirements.

Benefits of Requirements Management	
Benefit	**Explanation**
Better Change Control	Changes are reviewed before implementation.
Improved Communication	All stakeholders know about approved changes.
Reduced Project Risk	Prevents unexpected modifications.
Better Traceability	Every requirement can be tracked throughout the project.
Easier Maintenance	Future developers understand previous changes.
Higher Software Quality	Ensures software remains consistent with stakeholder needs.

Types of Software Requirements

Requirement Type	Description	Library Management System Example
Business Requirements	Describe the organization's goals and why the system is needed.	The university wants to automate library operations to reduce manual work and improve efficiency.
User Requirements	Describe what users need from the system in simple, non-technical language.	Students want to search for books online and renew borrowed books.
Functional Requirements (FR)	Specify what the system shall do.	The system shall allow students to search, borrow, return, and reserve books.
Non-Functional Requirements (NFR)	Specify how well the system should perform.	The system shall display search results within 2 seconds and be available 99.9% of the time.
Domain Requirements	Requirements that come from the application's domain, business rules, or regulations.	A student may borrow a maximum of 5 books according to university library policy.